

Statistical tests and assumptions

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Effect size and practical importance

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling variability

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Baselines and fair comparisons

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Statistical tests and assumptions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Statistical tests and assumptions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Differences₂, 2, 2 under the same null. Compute two-sided exact p and interpret.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Differences₁, 1, 1 have observed mean₁. Enumerate₈ assignments. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: The test requires its sign-exchangeability and independence assumptions; do not apply it blindly to clustered or asymmetric differences. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Differences2, 2, 2 under the same null. Compute two-sided exact p and interpret.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled statistical tests and assumptions evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of statistical tests and assumptions. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Statistical tests and assumptions

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

A times [5,8,4] seconds; B [4,6,3]. Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

Three independent paired task times

	A seconds	B seconds	B-A seconds
Task 1	5	4	-1
Task 2	8	6	-2
Task 3	4	3	-1

Read a cell using both its row and column labels.

Statistical tests and assumptions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preserve paired differences

Differences are -1,-2,-1; their sum is -4 and mean is $-4/3$ seconds.



2. Enumerate null sign flips

Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.



3. Count equally extreme sums

Only -4 and 4 have absolute sum at least 4: exact two-sided $p=2/8=0.25$.

CHANGE ONE THING

Break the task pairing and use unrelated A and B measurements.

Explain the changed result using the matching lesson.

Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p , and explain the decision at threshold 0.05.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only -4 and 4 have absolute sum at least 4: exact two-sided $p=2/8=0.25$.

1. Enumerate null sign flips
2. Count equally extreme sums
3. Preserve paired differences

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Count equally extreme sums Only -4 and 4 have absolute sum at least 4: exact two-sided $p=2/8=0.25$. A test compares an observed statistic with outcomes under a specified null model. Here the sign-flip assumptions make eight arrangements equally likely. At a predeclared 0.05 threshold, $p=0.25$ does not reject the null; it does not prove no difference.

Practice 2: explain the deciding evidence

Changed-case practice: Break the task pairing and use unrelated A and B measurements. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This paired sign-flip calculation is no longer justified by the stated design.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: This paired sign-flip calculation is no longer justified by the stated design. Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

Repair a missing building block

Review Effect size and practical importance (ai-7/13). Goal: Explain the mechanism and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the mechanism and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the mechanism and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
